Sustainable Tech Innovation

E-Revive

Smart E-Waste Second-Life Platform

Transforming discarded electronics into second-life assets through automated valuation, refurbishment, parts recovery, and certified recycling.

Market Challenge

1. The Problem Statement

A massive volume of functional or repairable electronic devices is dumped prematurely due to a lack of structured assessment and convenient disposal avenues.

Discarded Value & Environmental Risk

- **Premature Disposal:** Millions of smartphones, laptops, and chargers are thrown away even when minor repairs can restore full utility.
- **Lack of Consumer Awareness:** Device owners don't know where to dispose of e-waste safely or how to estimate residual worth.
- **Lost Resource Value:** Precious metals and reusable modules (RAM, displays, batteries) are permanently lost in landfills.
- **Informal Hazard:** Unregulated disposal leads to severe toxic soil and water contamination.



2. Proposed Idea: E-Revive

Smart Digital & Logistics Ecosystem

E-Revive is an intelligent digital platform that connects device owners with a circular second-life network.

Instead of routing all collected electronics straight to destruction, E-Revive assesses each device's condition to unlock its highest economic and functional value.

> **Core Objective:**
> Old Device → Smart Assessment → Repair/Reuse → Parts Harvest → Certified Recycling



4-Tier Device Classification

TIER 1

🟢 Direct Reuse

Device is in functional condition. Cleaned, data-wiped, and directed to secondary resale or donation channels.

TIER 2

🔵 Refurbish

Minor mechanical or battery repairs restore full utility, unlocking maximum resale margin for affordable tech buyers.

TIER 3

🟡 Parts Harvest

Non-repairable units are disassembled for usable displays, RAM modules, ICs, and copper wiring for repair markets.

TIER 4

🔴 Eco Recycling

Remaining hazardous materials & residue are safely handed over to CPCB-authorized metal extraction recyclers.

3. How the Solution Works

1. Register Device

User inputs specs, age, and condition via web or mobile app.

2. Smart Valuation

Algorithmic valuation generates immediate buyback offer.

3. Pickup & Triage

Doorstep collection followed by diagnostic testing.

4. Optimal Route

Device enters Reuse, Refurbishment, Harvesting, or Recycling.

4. Target Audience & Customers

Primary Consumers

- **College Students:** Seek affordable refurbished laptops and easy sell options.
- **Households:** Convenient doorstep disposal for old phones, chargers, and gadgets.
- **Individual Professionals:** Instant valuation and buyback rewards for upgrades.

B2B & Institutional

- **IT Companies & Offices:** Bulk e-waste retirement with compliance certificates.
- **Schools & Universities:** Subscription-based bulk pickup & recycling drives.
- **Authorized Recyclers:** Verified, steady supply of sorted e-waste residue.

5. What Makes E-Revive Novel?

TRADITIONAL E-WASTE FLOW

Collection Drive ➔ Direct Shredding & Recycling

(Recycling is the 1st option — destroys working components)

E-REVIVE CIRCULAR PARADIGM

Assess ➔ Reuse ➔ Refurbish ➔ Parts Harvest ➔ Recycle

(Recycling is the FINAL option — maximizes economic & component life)



6. Diversified Monetization Streams

1. Refurbished Sales

Buy used laptops (~₹3,000), repair (~₹1,000), sell (~₹5,000). High gross margins per restored device.

2. Parts Marketplace

Selling extracted functional RAM, displays, adapters, and circuit boards to repair shops.

3. Recycler Fee

Earning collection and sorting commissions from CPCB-certified scrap processors.

4. Enterprise B2B

Monthly compliance subscriptions for corporate e-waste auditing & disposal reports.

7. Financial Projections (~300 Units/Mo)

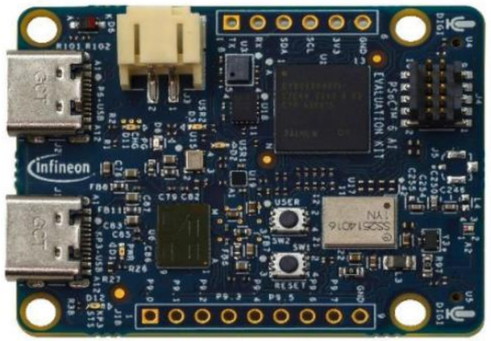


Estimated Gross Revenue: ~₹1,20,000/month. Estimated initial net profitability ranges between **₹30,000 – ₹50,000/month** after logistics, testing, and operational expenses.

8. Technology Stack & AI Integration

- **Frontend:** Responsive UI built with React.js, HTML5, and Tailwind CSS.
- **Backend & DB:** Node.js / Express microservices backed by MongoDB.
- **AI/ML Valuation Engine:** Predicts repairability probability and estimates market resale price based on specs, battery wear, and age.

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Input: [Device Age + RAM + Battery Health + Screen Condition]
→ ML Yield Prediction: ₹4,500 Resale Value
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Second Life for Electronics

Thank You!

"E-Revive turns discarded electronics into second-life products by intelligently deciding whether each device should be reused, refurbished, harvested for parts, or responsibly recycled."

Questions & Discussion

www.e-revive.platform contact@e-revive.org

Image Sources



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